Review for Dual hash: A Stochastic Primal-Dual Algorithm with Theoretical Guarantee for Deep Hashing

The paper presents DualHash, a primal-dual reformulation for deep hashing with W-type quantization regularization. My understanding is that the main contribution is an adaptation of primal-dual splitting, combined with stochastic momentum/STORM updates, to deep hashing. The method uses the Fenchel conjugate of a specific W-type regularizer to obtain a closed-form dual proximal update. However, the convexity of the dual/conjugate side is standard. The key question is whether this transformed problem provides meaningful guarantees for the original nonconvex W-regularized hashing objective. I do not find this connection convincingly established. In particular, the paper should clarify whether there is strong duality, zero duality gap, or at least a rigorous stationarity relationship between the transformed primal-dual problem and the original objective. I also found the setting and writing unclear. The term "W-type regularizer" is used repeatedly but never properly defined; only one example is given. Even by the start of Section 3, the precise problem being studied is still not fully clear. If "W-type" denotes a class of regularizers, that class should be formally specified. If it refers only to the particular regularizer $h(z) = \lambda \||z| - 1\|_1$, the paper should state this much earlier and avoid presenting the results as applying to a broader family.

Finally, the paper feels longer than necessary relative to its main contribution. Some of the space spent on standard preliminaries and routine descent arguments could be shortened, cited, or moved further into the supplement. This would allow the main text to focus more sharply on what is actually novel, especially the relationship between the primal-dual reformulation and the original nonconvex W-regularized hashing objective.